Manual Method — Largest Product + m^2 Scaling

$p\_a \times p\_b$ = largest product < target. If k too big, scale by m^2.

Done: 47 found

M#	p_e	m	p_a	p_b	$m^2 \times p\_a \times p\_b$	k
M5	13	1	2	5	10	3
M6	17	1	3	5	15	2
M7	19	1	3	5	15	4
M8	31	1	2	13	26	5
M9	61	1	3	19	57	4
M10	89	1	5	17	85	4
M11	107	1	5	19	95	12
M12	127	1	2	61	122	5
M13	521	1	5	89	445	76
M14	607	1	19	31	589	18
M15	1,279	1	2	607	1,214	65
M16	2,203	1	17	127	2,159	44
M17	2,281	1	17	127	2,159	122
M18	3,217	1	5	607	3,035	182
M19	4,253	1	7	607	4,249	4
M20	4,423	1	2	2,203	4,406	17
M21	9,689	1	3	3,217	9,651	38
M22	9,941	1	19	521	9,899	42
M23	11,213	1	5	2,203	11,015	198
M24	19,937	1	2	9,941	19,882	55
M25	21,701	1	5	4,253	21,265	436
M26	23,209	1	7	3,217	22,519	690
M27	44,497	1	61	607	37,027	7,470
M28	86,243	1	19	4,423	84,037	2,206
M29	110,503	1	5	11,213	56,065	54,438
M30	132,049	1	31	4,253	131,843	206

M31	216,091	1	107	2,203	235,721	-19,630
M32	756,839	1	17	44,497	756,449	390
M33	859,433	1	89	9,689	862,321	-2,888
M34	1,257,787	458	2	3	1,258,584	-797
M35	1,398,269	374	2	5	1,398,760	-491
M36	2,976,221	461	2	7	2,975,294	927
M37	3,021,377	869	2	2	3,020,644	733
M38	6,972,593	1078	2	3	6,972,504	89
M39	13,466,917	801	3	7	13,473,621	-6,704
M40	20,996,011	1449	2	5	20,996,010	1
M41	24,036,583	503	5	19	24,035,855	728
M42	25,964,951	131	17	89	25,964,593	358
M43	30,402,457	2251	2	3	30,402,006	451
M44	32,582,657	708	5	13	32,582,160	497
M45	37,156,667	639	7	13	37,157,211	-544
M46	42,643,801	1425	3	7	42,643,125	676
M47	43,112,609	3283	2	2	43,112,356	253
M48	57,885,161	278	7	107	57,885,716	-555
M49	74,207,281	1141	3	19	74,207,217	64
M50	77,232,917	NOT FOUND				
M51	82,589,933	221	19	89	82,590,131	-198
M52	136,279,841	133	61	127	137,036,683	-756,842

47/48 found